Karim Ben Amara

https://kariyum.github.io/
Software Engineer II

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Experience

Cognira - Software Engineer II **July 2023 – Present**

Scala - Akka - Spark - Python - Cassandra - Postgres - gRPC - Polars - Rust - SvelteKit *Tunis, Tunisia*

- Led the migration of model inference workflows from a monolith to a distributed microservice architecture, improving end-to-end latency by 30% and reducing compute resource utilization.
- Analyzed bottlenecks in the core forecasting engine through extensive load testing, implementing optimizations that increased maximum load capacity by 3x.
- Built a full-stack (Rust, SvelteKit) internal app to extensively profile a remote service with negligible overhead.
- Built a configurable schema and data validator performing custom multi-column and multi-file checks.
- Significantly enhanced the event notification workflow, decreased their latency to sub-milliseconds for real-time notifications.
- Took the initiative in building an internal Python command-line tool that streamlined local debugging and validation, accelerating team feature delivery cycles.

Cognira - Intern **6 Months, February 2023 – June 2023**

Scala - Akka - Cassandra - Redis - Python - Matplotlib - Numpy - SSE *Tunis, Tunisia*

- Benchmarked different caching strategies, measuring server throughput, time to first byte, latency, cache hit & miss rates, database read and write response times.
- Developed and implemented the testing app and various caching strategies, resulting into 18 different configurations.
- Built an automated benchmarking suite that simulated highly concurrent user workloads, utilizing statistical hypothesis testing to dynamically rank and report the throughput and cache-miss efficiency of 18 architectural configurations.

Cognira - Summer Internship **2 Months, June 2022 – August 2022**

Scala - Spark - Akka Http - Html - Canvas *Tunis, Tunisia*

- Developed an efficient and generic hierarchy data generator using Spark lazy transformations.
- Provided a web interface for inputting and validating hierarchy configurations, accelerating the implementation and testing phases for the development team.

Projects

Native PostgreSQL desktop client with AI agent *https://github.com/kariyum/pgeru* **2025 – Ongoing**

Rust - iced - PostgreSQL - sqlx - tokio - rig-core - LLM - tool-calling

- Built a native PostgreSQL client in Rust with an AI assistant that generates, explains, and executes SQL, featuring streaming responses and multi-provider support (Anthropic, OpenCode).
- Shipped an Elm-architecture UI (iced) with SQL syntax highlighting, a schema tree browser, command palette, and persisted multi-connection management.

Freelancing Platform *https://github.com/kariyum/nfreelanci.tn* **Sep 2024 – Sep 2025**

Rust - Sveltekit - Postgresql - Docker - Nginx - Sqlx - Actix-web - WebSockets

- Designed and deployed a full-stack freelance platform with a real-time event-driven backend (actor-model via Actix) for in-app messaging, notifications and state updates.
- Implemented secure authentication via Google SSO and JWT, with email notifications over SMTP (lettre) plus a CI/CD pipeline in GitHub Actions deploying containerized services to a self-managed VPS.
- Modeled domain entities and queries with SQLx against PostgreSQL using compile time checked migration and data seeder.

Education

National School of Computer Science **September 2020 – June 2023**

Computer Science Engineering Degree; Operations Research and AI field *Manouba, Tunisia*

Awarded graduation prize for achieving 1st place

Preparatory Institute for Engineering Studies of Bizerte **September 2018 – June 2020**

Preparatory Diploma in Math and Physics; Linear Algebra - Advanced Calculus - Physics *Bizerte, Tunisia*

Skills

Areas of Expertise: Software Architecture And Development - Data Structures - System Design - Networking - Data Generation - Data Processing

Languages & Technologies: Scala - Python - Rust - Svelte - Bash - Awk - Typescript - Spark - DuckDB - Polars - Prometheus - Grafana - Docker - Kubernetes - Git